

Prayas JEE 3.0 2025

Mathematics

DPP: 2

Inverse Trigonometric Functions

Q1 The range of function

$$f(x) = 2 \tan^{-1}(1 - x^2) + \left(\frac{\pi}{6}\right) \text{ is:}$$

- (A) $[-5\pi/6, 2\pi/3]$
 (B) $(-5\pi/6, 2\pi/3)$
 (C) $(-5\pi/6, 2\pi/3]$
 (D) $[-5\pi/6, 2\pi/3)$

Q2 Find x if $\sin^{-1} x - \cos^{-1} x = \cos^{-1} \frac{\sqrt{3}}{2}$

- (A) $\sqrt{3}$
 (B) $\sqrt{3}/2$
 (C) $-\sqrt{3}/2$
 (D) $2/\sqrt{3}$

Q3 The value of $\sin(2 \cos^{-1} x + \sin^{-1} x)$ when $x = \frac{1}{5}$

- (A) $1/5$ (B) $-1/5$
 (C) 5 (D) -5

Q4 If $a \sin^{-1} x - b \cos^{-1} x = c$, then $a \sin^{-1} x + b \cos^{-1} x$ is equal to:

- (A) 0
 (B) $\frac{\pi ab + c(b-a)}{a+b}$
 (C) $\frac{\pi}{2}$
 (D) $\frac{\pi ab + c(a-b)}{a+b}$

Q5 If $\tan^{-1} x + 2 \cot^{-1} x = \frac{2\pi}{3}$ then x is equal to

- (A) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$
 (B) 3
 (C) $\sqrt{3}$
 (D) $\sqrt{2}$

Q6

If $\tan^{-1}(\sin^2 \theta - 2 \sin \theta + 3) + \cot^{-1}$, then

$$\left(5^{\sec^2 y} + 1\right) = \frac{\pi}{2}$$

the value of $\cos^2 \theta - \sin \theta$ is equal to:

- (A) 0 (B) -1
 (C) 1 (D) None of these

Q7 If $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \dots\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots\right) = \frac{\pi}{2}$, for $0 < |x| < \sqrt{2}$, then x equals:

- (A) $\frac{1}{2}$
 (B) 1
 (C) $-1/2$
 (D) -1

Q8 The solution of the inequality $(\cot^{-1} x)^2 - 5 \cot^{-1} x + 6 > 0$ is:

- (A) $(\cot 3, \cot 2)$
 (B) $(-\infty, \cot 3) \cup (\cot 2, \infty)$
 (C) $(\cot 2, \infty)$
 (D) None of these

Q9 The value of ' a ', for which $ax^2 + \sin^{-1}(x^2 - 2x + 2)$ has a real solution $+ \cos^{-1}(x^2 - 2x + 2) = 0$ is:

- (A) $\frac{\pi}{2}$
 (B) $-\frac{\pi}{2}$
 (C) $\frac{2}{\pi}$
 (D) $-\frac{2}{\pi}$

Q10 The number of triplets (x, y, z) satisfying $\sin^{-1} x + \cos^{-1} y + \sin^{-1} z = 2\pi$, is:



(A) 0

(B) 2

(C) 1

(D) Infinite



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Answer Key

Q1 (C)

Q2 (B)

Q3 (A)

Q4 (D)

Q5 (C)

Q6 (C)

Q7 (B)

Q8 (B)

Q9 (B)

Q10 (C)



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